

ISACHSEN, NU, Canada

WMO: 710740

Lat: 78.792N

Lon: 103.554W

Elev: 58

StdP: 100.63

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.8	-42.3	-48.2	0.0	-43.7	-46.7	0.0	-42.2	21.6	-26.2	19.4	-27.2	1.9	220	1.226

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	12.0	8.4	10.0	7.2	8.1	6.1	8.6	11.7	7.4	9.9	6.2	8.2	4.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.3	6.0	9.7	5.2	5.5	8.6	4.2	5.1	7.4	26.2	12.0	23.7	10.0	21.1	8.3	12.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.3	12.7	DB	-45.9	12.5	2.6	4.0	-47.7	15.4	-49.2	17.8	-50.7	20.1	-52.5	23.0	(4)
(5)			WB	-45.4	9.0	0.4	2.2	-45.7	10.6	-45.9	11.8	-46.2	13.0	-46.5	14.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.8	-32.7	-33.1	-31.8	-22.7	-10.2	0.0	3.8	0.7	-7.0	-16.9	-23.6	-29.4	(6)	
	DBStd	14.19	4.63	5.25	5.79	4.87	4.94	3.69	3.47	3.32	4.32	5.68	5.53	5.54	(7)	
	HDD10.0	9786	1323	1206	1295	980	625	299	194	289	511	834	1007	1223	(8)	
	HDD18.3	12824	1581	1439	1553	1230	883	549	450	547	761	1092	1257	1481	(9)	
	CDD10.0	3	0	0	0	0	0	1	3	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	5.1	5.3	4.7	4.6	4.5	5.4	5.5	5.4	5.3	5.3	5.0	5.1	5.5	(14)	
(15) Precipitation	PrecAvg	125	4	4	4	5	12	13	22	24	18	13	4	4	(15)	
	PrecMax	207	14	16	20	14	28	33	53	50	61	26	10	13	(16)	
	PrecMin	69	0	0	0	0	2	2	4	8	1	1	0	0	(17)	
	PrecStd	37	4	4	5	5	6	9	15	12	16	7	3	4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.5	-17.9	-13.9	-9.9	1.5	12.9	15.3	12.0	2.4	-2.8	-7.3	-12.5	(19)	
		MCWB	-19.8	-18.3	-14.2	-10.5	-0.7	9.2	9.6	9.0	1.5	-2.9	-7.5	-12.8	(20)	
	2%	DB	-21.7	-21.3	-18.3	-12.4	-0.8	9.4	13.2	9.3	0.9	-4.6	-10.9	-15.9	(21)	
		MCWB	-22.0	-21.4	-18.6	-12.8	-1.6	6.6	8.9	7.0	0.2	-4.7	-11.3	-16.1	(22)	
	5%	DB	-23.8	-22.9	-21.4	-14.2	-2.3	7.0	11.3	7.2	-0.3	-6.3	-13.3	-18.5	(23)	
		MCWB	-24.0	-23.1	-21.6	-14.5	-2.8	4.7	7.9	5.9	-0.7	-6.5	-13.6	-18.7	(24)	
	10%	DB	-25.9	-24.8	-23.9	-16.2	-3.6	5.1	9.7	5.5	-1.4	-8.4	-15.6	-21.8	(25)	
		MCWB	-26.1	-25.0	-24.1	-16.5	-4.0	3.4	7.0	4.5	-1.6	-8.6	-15.8	-22.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-19.8	-17.8	-13.9	-10.3	0.0	9.2	10.2	9.5	1.7	-2.8	-8.3	-12.6	(27)	
		MCDB	-19.6	-17.4	-13.6	-10.1	1.4	12.2	14.4	12.0	2.2	-2.8	-8.0	-12.4	(28)	
	2%	WB	-22.8	-21.4	-18.3	-12.6	-1.5	6.7	9.1	7.3	0.5	-4.6	-12.6	-16.9	(29)	
		MCDB	-22.6	-21.2	-18.1	-12.4	-0.8	9.4	12.7	9.0	0.9	-4.5	-12.3	-16.7	(30)	
	5%	WB	-24.6	-23.3	-21.9	-14.5	-2.8	5.0	8.1	6.1	-0.5	-6.3	-15.0	-20.0	(31)	
		MCDB	-24.4	-23.1	-21.7	-14.2	-2.3	6.9	11.2	7.5	-0.2	-6.1	-14.7	-19.8	(32)	
	10%	WB	-27.0	-25.8	-24.6	-16.5	-4.0	3.5	7.2	4.7	-1.4	-8.4	-17.3	-22.6	(33)	
		MCDB	-26.9	-25.7	-24.5	-16.3	-3.7	5.1	9.7	5.7	-1.2	-8.3	-17.0	-22.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.6	5.9	6.1	4.5	3.4	4.3	3.6	4.5	5.7	5.8	5.7	(35)	
		MCDBR	7.2	7.2	8.5	7.3	5.0	5.6	6.8	5.9	4.2	7.3	7.4	8.5	(36)	
		MCWBR	7.5	8.0	8.8	7.3	4.2	3.8	4.3	4.2	3.8	7.5	8.1	9.5	(37)	
	5% WB	MCDBR	7.6	8.0	8.8	7.3	4.4	5.7	6.7	5.7	3.8	7.1	7.3	9.3	(38)	
		MCWBR	7.5	7.9	8.8	7.3	3.9	3.9	4.4	4.2	3.6	7.2	7.2	9.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		N/A	N/A	0.210	0.276	0.302	0.329	0.342	0.297	0.247	N/A	N/A	N/A	(40)	
	taud		N/A	N/A	2.059	2.065	2.092	2.148	2.248	2.358	2.116	N/A	N/A	N/A	(41)	
	Ebn at Noon		0	0	607	753	807	799	767	748	548	0	0	0	(42)	
	Edh at Noon		0	0	46	90	108	109	94	69	46	0	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.00	0.00	0.71	3.46	5.38	6.34	5.35	2.91	0.99	0.00	0.00	0.00	(44)	
	RadStd		0.00	0.00	0.08	0.23	0.42	0.62	0.47	0.29	0.09	0.00	0.00	0.00	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page